1.02 Surds

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Recap:

Note: A *rational* number is any which can be expressed as $\frac{a}{b}$ where a, b are integers. $\frac{2}{3}$ and $\frac{4}{1} = 4$ are rational numbers, but π and $\sqrt{2}$ are not.

A surd is a root of a number that does not simplify to a rational number.

Laws:

$\sqrt{a} \times \sqrt{b} = \sqrt{ab}$

$\frac{\sqrt{a}}{\sqrt{b}} = \sqrt{\frac{a}{b}}$

$\sqrt{3} \times 2$

$3\sqrt{5} \times 2\sqrt{5}$

$\sqrt{8} = \sqrt{4}\sqrt{2}$

$\sqrt{12} + \sqrt{27}$

$\left(\sqrt{8} + 1\right)\left(\sqrt{2} - 3\right)$

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3 Simplify $\sqrt{75} - \sqrt{12}$ giving your answer in the form $a\sqrt{3}$, where a is an integer.

Extension

1

[SMC 2014 Q24] Which of the following is smallest?

$10 - 3\sqrt{11}$; $8 - 3\sqrt{7}$; $5 - 2\sqrt{6}$; $9 - 4\sqrt{5}$; $7 - 4\sqrt{3}$.

2

[SMC 2012 Q21] Which of the following numbers does not have a square root in the form $x + y\sqrt{2}$, where x and y are positive integers?

$17 + 12\sqrt{2}$; $22 + 12\sqrt{2}$; $38 + 12\sqrt{2}$; $54 + 12\sqrt{2}$; $73 + 12\sqrt{2}$.

1.02 Surds

Chapter 1 · Original slide 24 · 6: Rationalising The Denominator

Here's a surd. What could we multiply it by such that it's no longer an irrational number?

$\sqrt{5}\times$

$\frac{1}{\sqrt{2}}$

In this fraction, the denominator is irrational. 'Rationalising the denominator' means making the denominator a rational number.

What could we multiply this fraction by to both rationalise the denominator, but leave the value of the fraction unchanged?

Side Note: There's two reasons why we might want to do this:

1. For aesthetic reasons, it makes more sense to say "half of root 2" rather than "one root two-th of 1". It's nice to divide by something whole!

2. It makes it easier for us to add expressions involving surds.

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$\frac{3}{\sqrt{2}} =$

$\frac{6}{\sqrt{3}} =$

$\frac{7}{\sqrt{7}} =$

$\frac{15}{\sqrt{5}} + \sqrt{5} =$

1.02 Surds

Chapter 1 · Original slide 26 · More Complex Denominators

You've seen 'rationalising a denominator', the idea being that we don't like to divide things by an irrational number.

We basically use the same expression but with the sign reversed (this is known as the *conjugate*). That way, we obtain the difference of two squares. Since $(a+b)(a-b) = a^2 - b^2$, any surds will be squared and thus we'll end up with no surds in the denominator.

But what do we multiply the top and bottom by if we have a more complicated denominator?

$\frac{1}{\sqrt{2}+1} \times$

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$\frac{3}{\sqrt{6}-2} \times$

You can explicitly expand out $\left(\sqrt{6}-2\right)\left(\sqrt{6}+2\right)$ in the denominator, but remember that $(a-b)(a+b) = a^2 - b^2$ so we can mentally obtain $6 - 4 = 2$

Just remember: 'difference of two squares'!

$\frac{4}{\sqrt{3}+1} \times$

$\frac{3\sqrt{2}+4}{5\sqrt{2}-7} \times$

1.02 Surds

Chapter 1 · Original slide 28 · Test Your Understanding

Rationalise the denominator and simplify

$\frac{4}{\sqrt{5}-2}$

Solve $y\left(\sqrt{3}-1\right)=8$

Give your answer in the form $a+b\sqrt{3}$ where a and b are integers.

Rationalise the denominator and simplify

$\frac{2\sqrt{3}-1}{3\sqrt{3}+1}$

Chapter 1 · Original slide 29 · Writing surd expressions in power form

This is not in the textbook, but a common type of question in exams, often asked in conjunction with differentiation (a later chapter).

Express $\frac{\left(2+\sqrt{x}\right)^2}{x}$ as powers of x.

Write any roots as powers of x.

Test Your Understanding

Express $\frac{\left(x+\sqrt{x}\right)^2}{2\sqrt{x}}$ as powers of x.

Expand.

Split fraction (some may wish to do this step mentally)

1.02 Surds

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or alternatively: (not in textbook)

1 a b c d e

Rationalise the denominator and simplify the following:

$\frac{1}{\sqrt{5}+2} =$

$\frac{\sqrt{3}}{\sqrt{3}-1} =$

$\frac{\sqrt{5}+1}{\sqrt{5}-2} =$

$\frac{2\sqrt{3}-1}{3\sqrt{3}+4} =$

$\frac{5\sqrt{5}-2}{2\sqrt{5}-3} =$

2 3 4 5 6

Expand and simplify:

$\left(\sqrt{5}+3\right)\left(\sqrt{5}-2\right)\left(\sqrt{5}+1\right) =$

Rationalise the denominator, giving your answer in the form a+b√3.

$\frac{3\sqrt{3}+7}{3\sqrt{3}-5} =$

Solve x(4−√6)=10 giving your answer in the form a+b√6.

Solve y(1+√2)−√2=3

Simplify:

$\frac{\sqrt{a+1}-\sqrt{a}}{\sqrt{a+1}+\sqrt{a}} =$